Enhancing Adaptive Teaching of Reading Skills Using Digital Technologies: The LATILL Project

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Abstract. Level-adequate reading materials are a key factor to successful foreign language learning. However, teachers need to rely either on prepackaged materials or their experience-based intuition to identify texts matching the required CEFR level. In this context, the LATILL platform aims to provide teachers of German as a foreign and second language with powerful tools for identifying level-adequate texts, helpful information on teaching reading and motivating materials for the classroom. The project will also provide a professional development programme to support level-and-learner-adequate teaching of reading skills. The main beneficiaries will be the learners of German as a foreign or second language, they will have access to up-to-date reading material that meets their specific needs and interests, which is expected to have a positive impact on their reading motivation as well as on their general proficiency in the long term.

Keywords: foreign language · German · Open Education · Natural Language Processing · knowledge management ecosystem

1 Introduction

Reading competences in foreign languages are essential for students to gain access to information in other countries, to compare different perspectives and to cooperate in international contexts [1]. Therefore, enhancing literacy is an important goal of foreign language teaching and several didactic approaches have been developed to this end [2–5]. Together with an adequate teaching methodology, taking into account learners' individual needs and interests, the selection of suitable reading material is central in order to foster reading competence. Appropriate texts should (1) address current issues

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to build and maintain reading motivation and (2) meet the language levels of students to develop their reading competences [4, 6]. In order to increase fluency in reading, it is of particular importance that the texts' linguistic features correspond to the individual learner's level of second language acquisition [7, 8].

The reading materials provided in textbooks, however, cannot fully meet these requirements: Texts become outdated very quickly and a range of different texts often is needed for heterogeneous groups of learners. On the other hand, up-to-date texts that could be found via content-based Internet searches mostly require advanced reading skills and are thus not suitable for learners on beginner or intermediate language level [9].

In this context, the LATILL project (Level-Adequate Texts in Language Learning) aims to provide open educational resources for teachers of German as a foreign language that meet the need for current, authentic and, most importantly, level-adequate reading materials (Table 1). Therefore, a freely accessible platform that offers a search and analysis function for German texts on a specific topic and CEFR level [1] is to be created. Going beyond language-aware search engines [10, 11], the platform will also provide auxiliary tools and materials for working with authentic texts to support teachers in implementing the developed tools and address the issue of teaching methodology.

To initiate change at the classroom level and to overcome theory-practice problems of top-down teacher education approaches, the LATILL platform will be developed in a joint effort of foreign language researchers, computer science researchers and teachers. This collaboration involves the creation of professional development modules for foreign language teachers that are grounded in experience-based and reflexive principles. An accompanying evaluation study will look into the implementation of the developed tools

Table 1. Project details.

Title	Level-Adequate Texts in Language Learning
Acronym	LATILL
Funding entity	European Union
Call	Erasmus + KA220-SCH - Cooperation partnerships in school education
Reference	2021–1-AT01-KA220-SCH-000029604
Project leader	Karen Schramm
Coordinator	Universität Wien (Austria)
Partners	Universidad de Salamanca (Spain) Eberhard Karls Universität Tübingen (Germany) Chernivtsi National University Yuriy Fedkovych (Ucraine) Verein Österreichisches Sprachdiplom Deutsch (Austria)
Budget	351.091 €
Start date	01/05/2022
End date	30/04/2025

and materials in Spanish and Ukrainian classrooms at the secondary level and allow for conclusions on desirable improvements of the tools, materials and accompanying measures.

This contribution presents the background (Sect. 1), the project consortium (Sect. 2), the objectives and work plan (Sect. 3) and the planned outcomes (Sect. 4) of the LATILL project. Finally, the last Sect. (5) summarises the main conclusions.

2 The Consortium

The project is coordinated by the University of Vienna. The consortium consists of five European institutions from four countries: four public Universities and one non-profit association, the state-approved examination and assessment institution Österreichisches Sprachdiplom Deutsch (ÖSD).

- Universität Wien (Austria),
- Universidad de Salamanca – represented by the GRIAL Research Group (Spain),
- Eberhard Karls Universität Tübingen – represented by the ICALL Research Group, http://icall-research.de (Germany),
- Chernivtsi National University Yuriy Fedkovych (Ukraine),
- Österreichisches Sprachdiplom Deutsch (ÖSD), https://www.osd.at, (Austria).

Moreover, several secondary schools in Spain and Ukraine are involved in the project as associated partners. These schools do not form part of the consortium, but are associated with the project and will take part in the professional development programme and the evaluation study of LATILL. Each secondary school involved signed a letter of intentions to agree on the tasks in which teachers will be involved across the project.

3 The Project

3.1 The Objectives

The main objective of the LATILL project is to promote up-to-date, learner-oriented and inclusive teaching of reading skills with a special focus on German as a foreign language. Therefore, a freely accessible open education platform will be developed that includes an innovative tool for text search and analysis as well as supporting materials and professional development modules.

The LATILL project aims to have a positive impact on language teaching and learning by promoting innovative teaching methods, expanding the range of open educational resources in line with the current reference levels of the CEFR, encouraging the use of up-to-date, learner-oriented materials (expected to have a positive impact on learner motivation) and promoting internal differentiation, considering different interests and needs of learners. Moreover, the tool for search and analysis of texts according to the current levels of the CEFR, along with supporting materials and training measures, enhances quality assurance in language teaching. With the implementation of the professional development programme, LATILL also aims to promote cooperation and exchange of

teaching materials among teachers at national and international levels and to strengthen the profile of the teaching profession through further training and participation in an international research project.

3.2 The Work Plan

The project goals are divided into the following five intellectual results (Fig. 1):

- Development and maintenance of an open education platform to provide tools and didactic material for teaching reading skills (R1).
- Supporting adaptive retrieval of attractive reading material using topic- and CEFR-classification (R2).
- Development of didactic materials to support level-and-learner-adequate teaching of reading skills (R3).
- Concept for and implementation of a professional development programme to support level-and-learner-adequate teaching of reading skills and evaluation of the implementation of the developed tools and materials in the classroom (R4).
- Creation of a CEFR Reference Corpus for German (R5).

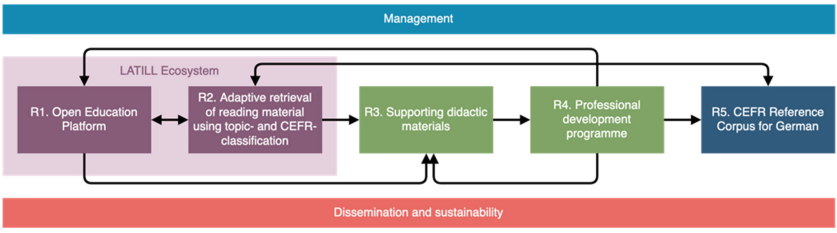


Fig. 1. Relationships between the project results

4 Project Results

4.1 Open Education Platform (R1)

The main objective of this output is to develop an open education platform to provide the necessary mechanisms to support German language teachers to implement innovative teaching methods using digital technologies and open educational resources (OER) [12]. The platform will support the didactical approach defined in R3 and it will be integrated with the tool developed in R2. The connection between R1 and R2 will provide the ecosystem [13, 14] to support teachers in implementing uniform quality standards in foreign language teaching. The platform will also provide tools for teachers to analyse their own German text, although these texts will not be included in the corpus in order to avoid legal problems related to copyrights.

The platform will be developed using Open Source software to ensure the sustainability of results and facilitate the integration into the technological ecosystems of the schools. The development of the platform will follow a user-centred approach. Final users (teachers) and experts will be involved in the design of the LATILL platform in order to best determine the constraints that must be used in developing the platform.

4.2 Adaptive Retrieval of Reading Material Using Topic- and CEFR-Classification (R2)

The goal of this result is to support the adaptive identification of texts that are both of interest to the reader and at the appropriate proficiency level. The texts retrieved should be attractive to read at the given proficiency level and rich in language forms fostering learning at the targeted level.

For identifying texts of interest, the Machine Learning (ML) models will be developed to support a characterisation in terms of the general topic (e.g., health, education, politics, culture, history, nature, business, sports, travel) as well as specific search queries for a more focused characterisation. The automated complexity analysis will support a characterisation of texts in terms of the standard CEFR levels (R5), while also permitting the (de)prioritisation of particular language constructions to make it possible to identify texts that are rich in pedagogically targeted, developmentally proximal language constructions. ML model performing the CEFR-qualification will be calibrated in relation to the top-down characterisation developed by Profile Deutsch and validated bottom-up against the CEFR reference corpus that project result 5 provides and ensures validity for through standard setting. Further levelled reading material resources will be employed to ensure appropriate cross-corpus generalizability of the CEFR classification model.

The approach will support both retrieval of texts using a web search engine with the computational analysis performed dynamically, as well as pre-analysed web-sources that were identified by the partners as particularly relevant (henceforth: LATILL sources) for which the linguistic complexity and topic analyses can be precomputed to facilitate efficient filtering and ranking providing systematic access to didactically and thematically attractive materials.

4.3 Supporting Didactic Materials (R3)

Freely accessible didactic materials will be created to support teachers in the use of the LATILL platform and tools and to enhance the use of up-to-date, learner-orientated teaching methodology. These materials include instruction manuals for the platform and especially the text retrieval tool, guidelines for learner-orientated text selection and methodical recommendations for working with authentic texts. The information will be available in German, Spanish and Ukrainian.

4.4 Professional Development Programme (R4)

A professional development programme for German-as-a-Foreign-Language teachers that focuses on level-and-learner-adequate teaching of reading skills will be conceptualised, implemented and evaluated in the context of secondary school teaching in Spain

and Ukraine. The programme consists of a 3-day online workshop organised at the University of Vienna and four online learning modules. The first part of the training focuses on the participants' prior knowledge, reflection about reading skills training and the introduction to the LATILL platform and tools. In the second part, teachers get the chance to expand their methodical competence, get to know new features of the platform and reflect on the use of the LATILL materials. In addition, an evaluation study will be conducted to ensure the usability of the developed tools and materials as well as the perceived usefulness of the teacher training. This study is based on focus groups and complements the usability study that forms part of project result 1.

4.5 CEFR Reference Corpus for German (R5)

The goal of the last result is to establish a reference data set that can be used to train the ML model performing the CEFR-classification of the reading material. This is required to ensure that the texts retrieved by the platform are globally appropriate at the specified proficiency level and rich in language forms fostering learning at the targeted level, as ensured by the linguistic complexity analysis resulting from project result 2. The CEFR reference here serves to provide the data-driven complement of the top-down characterisation of the relevant language aspects characterised by Profile Deutsch.

5 Conclusions

The LATILL project is focused on improving teaching practices related to German as a foreign or second language. In particular, the project aims to promote the use of up-to-date, learner-oriented materials in the classroom; to encourage teachers to apply innovative teaching methods, including the use of digital technologies; to promote the implementation of uniform quality standards in foreign language teaching, and to encourage cooperation and exchange among teachers at national and international level.

The LATILL platform will provide a search and analysis function for German texts on a specific topic and CEFR level as well as supporting tools and didactic materials for teachers. Teachers from secondary schools will use the platform in their classrooms following an open educational approach. Furthermore, although the professional development program and the classroom implementation (R4) focus on the secondary school context, the LATILL tools and materials are also valuable in the field of higher education and adult education (German classes at university level, public and private language schools, job-related German courses etc.).

Learners of German as a foreign or second language will especially benefit from being exposed to up-to-date reading material that meets their specific needs and interests, which is expected to have a positive impact on their reading motivation as well as on their general proficiency in the long term.

Regarding the risks associated with this project, we mainly identified issues related to changes inside the secondary schools due to COVID-19. Nowadays, an innovative approach to changing teaching practices must consider different scenarios, not only face-to-face lessons. The LATILL platform (R1 and R2) and the didactic approach (R3) will be compatible with blended and online learning. Moreover, the programme to support

level-and learner-adequate teaching of reading skills (R4) is based on online learning modules, so the training activities can be organised fully online if needed.

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